

§11.4 - The Comparison Tests

Lemma #1

Let (a_n) be a sequence of nonnegative numbers.

Then

1. the series $\sum a_n$ converges to a finite number,
2. or the series $\sum a_n$ diverge to infinity.

Theorem #1 (Basic Comparison Test)

Let (a_n) and (b_n) be sequences such that

$$0 < a_n \leq b_n$$

for all $n \in \mathbf{N}$.

Then

$$\sum a_n = \infty \quad \text{implies} \quad \sum b_n = \infty$$

and

$$\sum b_n \in \mathbf{R} \quad \text{implies} \quad \sum a_n \in \mathbf{R}.$$

Remark

The condition

$$0 < a_n \leq b_n \text{ for all } n$$

in the Basic Comparison Test can be relaxed to hold for all n -large.

The condition

$$f \text{ decreasing on } [1, \infty)$$

in the Integral Test can be relaxed to f decreasing for all x -large.

Example #1

Determine if

$$\sum_{k=2}^{\infty} \frac{1}{\sqrt{k} - 1}$$

converges.

Example #2

Determine if

$$\sum_{k=2}^{\infty} \frac{\ln(k)}{k^2}$$

converges.

Example #3

Determine if

$$\sum_{k=1}^{\infty} \frac{k}{k^2 + 2k + 3}$$

converges.

Theorem #2 (Limit Comparison Test)

Let (a_n) and (b_n) be sequences of positive numbers where

$$c = \lim_{n \rightarrow \infty} \frac{a_n}{b_n}.$$

exists.

If $c > 0$ and finite then

$$\sum a_n \quad \text{and} \quad \sum b_n$$

either both converge or both diverge.

Example #4

Determine if

$$\sum_{k=1}^{\infty} \frac{e^{2k} + k}{e^{5k} - k^2}$$

converges.

Example #5

Determine if

$$\sum_{k=1}^{\infty} \frac{\sin(1/k)}{k}.$$

converges.